CARE-Judge: Calibrated, Abstaining, and Robust Evaluation via Competence-Gated Protocol Stability for LLM-as-a-Judge

Anonymous submission

Abstract

Large language models are increasingly used as automated judges to evaluate generated text, yet they are overconfident, positionally biased, and sensitive to evaluation-protocol wording. We study *protocol (in)stability* as a per-instance reliability signal for LLM judges—whether a verdict survives semantically equivalent rubric paraphrases, response-order permutations, and stochastic resampling—and report that the signal is **competence-gated**: it predicts error above a judge-capability threshold and inverts below it. Above the threshold, instability tracks error (accuracy gaps up to 27 points between stable and unstable subsets on a mid-capability judge); below it, a near-random judge produces *stable-but-wrong* verdicts (65% of Qwen2.5-1.5B's JudgeBench errors are rubric-stable). A negative control supports the mechanism: on a mid-capability judge, intentionally non-equivalent rubrics flip verdicts $2.2\times$ more often than equivalent paraphrases, whereas a near-random judge is insensitive to the distinction. We operationalize this as **CARE-Judge** (Calibrated, Abstaining, and Robust Evaluation), which fuses protocol-stability features into a learned correctness calibrator with an inherited selective-risk threshold (an exact Clopper–Pearson bound, validated by measured violation rates rather than a new theorem). Across five judges (a Qwen2.5-1.5B/7B ladder, a controlled 1.5B/7B/14B size ablation, DeepSeek-V4, GPT-5.5) and four benchmarks (JudgeBench, TL;DR, RewardBench, LMArena; 22,000+ pairwise evaluations), fusion is positive on seven of the eight pre-specified mid-capability pairs (up to $+7.6$ AUROC over the best single signal; the exception is Qwen2.5-7B on JudgeBench), ties already-strong base confidence for the frontier judge, and adds nothing where a judge is near-chance. Pooling all 17 pairs dilutes this to a one-sided Wilcoxon p=0.087, as expected when regimes where fusion cannot help are averaged in. An anonymized code and feature-trace archive accompanies the submission.

1 Introduction

Large language models (LLMs) are now widely deployed as automated judges—scoring RLHF outputs (Ouyang et al. 2022), ranking chatbot responses (Zheng et al. 2023), and evaluating generation quality—but they make systematic errors, exhibit positional biases, and vary verdicts based on evaluation-protocol wording, all while expressing high confidence (Li et al. 2024; Wang et al. 2023; Wei et al. 2024).

Prior work controls this risk explicitly but through a *single* confidence source: Jung, Brahman, and Choi (2025) (Trust-or-Escalate) use simulated-annotator agreement with cascaded selective evaluation, and Badshah, Emami, and Sajjad (2026) (SCOPE) aggregate both response orderings into a permutation-invariant conformal score. Neither exploits the broader observation that sensitivity across *multiple* semantically equivalent evaluation protocols is itself an error signal. We build on their selective-risk machinery—a held-out Clopper–Pearson threshold, which we adopt rather than claim as a contribution (Section 2); because our threshold is chosen data-dependently, we treat the risk criterion as an empirical target validated by measured violation rates rather than a family-wise-corrected test.

We investigate a complementary and asymmetric hypothesis: **a judgment that is unstable under semantically equivalent evaluation protocols is often unreliable, whereas stability alone is not sufficient evidence of correctness.** Concretely, if an LLM judge produces different rankings when (i) the rubric is paraphrased, (ii) the response order is swapped, or (iii) the judgment is re-sampled at nonzero temperature, the resulting verdict is less trustworthy. The converse does not hold: a judge can be stably wrong, applying the same erroneous heuristic (e.g., a length or fluency preference) under every protocol variant (Xu et al. 2026). We call this property *protocol stability*, and we show that protocol *instability* is a useful warning signal where the judge is at least moderately accurate, and we identify the competence regime in which it fails.

We organize our study around three questions: *(RQ1)* does per-instance protocol stability predict judge error? *(RQ2)* does fusing protocol-stability signals improve the risk–coverage tradeoff over standard confidence and single-signal baselines at a matched budget? *(RQ3)* in which capability and task regimes do these signals help, and where do they break down?

Building on this insight, **CARE-Judge** (Calibrated, Abstaining, and Robust Evaluation; Figure 1) collects the three protocol-stability signal families per item, learns a correctness calibrator $\hat{p}(\text{correct} \mid \text{features})$ on a training split, selects an acceptance threshold on a disjoint calibration split via an empirical prefix search certified with an exact Clopper–

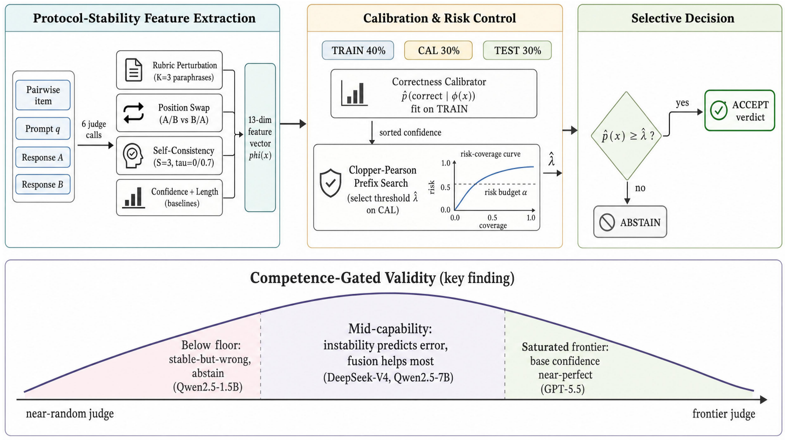

Figure 1: Overview of CARE-Judge. Six judge calls per item yield three protocol-stability feature families (rubric perturbation, position swap, self-consistency) plus confidence/length baselines ($\phi(x)$); a correctness calibrator fit on TRAIN and an exact Clopper–Pearson threshold $\hat{\lambda}$ selected on CAL drive an accept/abstain decision on held-out TEST. **Key finding (bottom):** the signal is *competence-gated*—it inverts below a competence threshold (abstain), helps most in the mid-capability band, and is redundant for a saturated frontier judge.

Pearson bound (Jung, Brahman, and Choi 2025; Clopper and Pearson 1934), and reports selective risk, coverage, and violation rates on a held-out test split.

We validate CARE-Judge across five judges spanning a wide capability range—two API judges (DeepSeek-V4, GPT-5.5) and a same-family Qwen2.5-1.5B/7B ladder evaluated on all four benchmarks, plus a *controlled* 1.5B/7B/14B size-ablation ladder on JudgeBench (Section 4.6) that isolates the parameter-count effect—over four benchmarks (JudgeBench (Tan et al. 2024), RewardBench (Lambert et al. 2024), TL;DR (Stiennon et al. 2020), LMArena (Zheng et al. 2023)) and 22,000+ pairwise evaluations. The throughline is that *the utility of protocol stability is governed by judge competence*: instability predicts error above a competence threshold (up to 27-point accuracy gaps, RQ1); fusion helps most in the mid-capability regime, improving seven of the eight pre-specified mid-capability pairs (up to $+7.6$ AUROC over the best single signal, on Qwen2.5-7B/RewardBench; up to $+6.3$ on DeepSeek-V4), while it adds nothing to a saturated frontier judge (RQ2); and below the threshold the signal *inverts* into stable-but-wrong verdicts CARE correctly abstains on (RQ3).

Our contributions are:

1. **The competence-gated asymmetry of protocol stability (primary).** Protocol instability predicts judge error only above a minimum judge-competence threshold; below it the signal *inverts*, and a near-random judge produces stable-but-wrong verdicts (Section 4.4). A negative-control experiment supports the mechanism: on a mid-capability judge, intentionally non-equivalent rubrics flip verdicts $2.2\times$ more often than equivalent paraphrases, whereas a near-random judge is insensitive to the distinction (Section 4.2), and a *controlled* same-family capability ladder traces the boundary. This asymmetry distinguishes our finding from prior per-instance judge-reliability work (Section 2) and explains why single-signal judge-reliability methods succeed on some judges and fail on others.
2. **A risk-controlled selective evaluator that operationalizes the finding.** We fuse three families of protocol-perturbation instability—rubric paraphrase, response permutation, and stochastic resampling—into a *single per-instance accept/abstain decision under an explicit risk budget*. Consistent with the competence-gating account, this fusion gives its largest gains in the mid-capability regime—positive on seven of the eight pre-specified mid-capability judge–benchmark pairs—above standard confidence, single-signal baselines (including the SCOPE-style bidirectional signal), and ensemble aggregation at a matched inference budget, while adding nothing where a frontier judge is already saturated or a weak judge is below the threshold (Section 4). Prompt- and order-sensitivity have been studied before as *diagnostic* properties (Hua et al. 2025; Guan et al. 2025; Anghel et al. 2025a; Huang, Sun, and Yang 2026), and individual consistency signals have been used to *rank* or ensemble judges (Li et al. 2025b; Gupta and Kumar 2026); our contribution is neither observing protocol sensitivity nor a single consistency check, but combining all three perturbation families into one calibrated, risk-controlled acceptance rule whose regime of validity we characterize (Section 3).

An anonymized code and feature-trace archive accompanies this submission as supplementary material, with the full public release to follow under the same license upon publication.

2 Related Work

LLM-as-a-Judge. Pairwise LLM judging is now a standard scalable evaluation paradigm (Zheng et al. 2023), and surveys document its position, length, and self-enhancement biases (Li et al. 2024); mitigations include swap-and-compare and debate (Wang et al. 2023), and prompt/rubric-template sensitivity is well established (Wei et al. 2024). Two recent findings directly motivate our design: Xu et al. (2026) give a mechanistic account showing judge biases arise from shared internal heuristics—so stability can be a warning signal but never a correctness guarantee, and verbalized confidence can be unfaithful—and Li et al. (2025a) document preference leakage, underscoring the need for strictly disjoint splits when calibrating any judge-reliability estimator.

Prompt-Sensitivity, Consistency, and Per-Instance Judge Reliability. A growing body of work studies how LLM outputs shift under semantically equivalent prompt, rubric, or ordering perturbations, and how such shifts bear on *per-instance* reliability. Hua et al. (2025) ask whether paraphrase sensitivity is a flaw or an artifact; Guan et al. (2025) isolate the input-order effect; Anghel et al. (2025a) build a pairwise meta-evaluator diagnosing judge instability; and Huang, Sun, and Yang (2026) use prompt perturbation to stabilize pairwise evaluation. Closer to the per-instance regime, Gupta and Kumar (2026) diagnose judge reliability with conformal prediction sets, Choi et al. (2026) apply item-response theory, and Li et al. (2025b) ensemble prompt variants to raise accuracy. Ling et al. (2025) caution that abstention can itself be a prompt artifact—a caveat we address via competence-gating (Section 4.4).

We do *not* claim to be the first to observe protocol sensitivity or to use a consistency signal; we differ in *use* and *scope*. In use, prior work diagnoses judges globally (Hua et al. 2025; Guan et al. 2025; Anghel et al. 2025a), builds per-instance reliability *sets* for analysis (Gupta and Kumar 2026; Choi et al. 2026), or ensembles prompts to *improve accuracy* (Li et al. 2025b; Huang, Sun, and Yang 2026); we instead fuse three perturbation families into a single calibrated accept/abstain decision under an explicit risk budget. In scope, none characterizes the competence-gated asymmetry we report—instability predicts error above a competence threshold and inverts below it.

Selective Prediction and Abstention. Selective classification (Geifman and El-Yaniv 2017; El-Yaniv and Wiener 2010) lets a predictor abstain on uncertain inputs; recent LLM work spans abstention surveys (Wen et al. 2025), uncertainty-calibrated fine-tuning (Krishnan, Khanna, and Tickoo 2024), selective conformal uncertainty with finite-sample error control (Wang et al. 2025), and learned conformal abstention policies (Tayebati et al. 2025). We differ by using *evaluation-protocol stability* as the uncertainty source, specific to judging and requiring neither model internals nor a learned controller.

Risk-Controlled Selective Judging. Our work is closest to a recent line of risk-controlled selective LLM judging, and we are explicit that the statistical machinery is adopted, not invented. Jung, Brahman, and Choi (2025) (Trust-or-Escalate) proposed cascaded selective evaluation with human-agreement guarantees using Simulated Annotators plus Clopper–Pearson risk control; concurrently Badshah, Emami, and Sajjad (2026) (SCOPE) aggregate both response orderings into a permutation-invariant conformal score, and Sheng et al. (2025) apply conformal prediction to judge-uncertainty intervals. CARE-Judge reuses this apparatus (an exact Clopper–Pearson threshold on a held-out split) rather than proposing new guarantees. Our contribution is the *confidence signal*: where Trust-or-Escalate relies on simulated-annotator agreement and SCOPE on bidirectional probability alone, we fuse rubric, position, and self-consistency instability into one calibrator and characterize the capability regimes where such signals help or fail—making CARE-Judge a multi-view generalization complementary to any single signal.

Judge Benchmarks and Rubric-Based Evaluation. We use JudgeBench (Tan et al. 2024) and RewardBench (Lambert et al. 2024) as hard, objectively-labeled benchmarks and TL;DR (Stiennon et al. 2020) and LMArena (Zheng et al. 2023) as subjective-preference benchmarks. Prior rubric work treats rubrics as fixed criteria to align with—Hong et al. (2026) align judges with human scoring rubrics and Anghel et al. (2025b) propose a rubric-driven multi-metric framework—whereas we treat *rubric sensitivity* (how much a verdict changes under semantically equivalent paraphrases) as an uncertainty signal. The specific combination we study—rubric-paraphrase instability as a per-instance acceptance signal within a risk-controlled selective-judging framework—is, relative to the work above, the narrower gap this paper addresses, and we validate it empirically.

Cost-Aware Cascades. Routing easy queries to cheap models and escalating hard ones is a well-established cost-reduction strategy; Zellinger, Liu, and Thomson (2025) study LLM cascades with early abstention and formal cost analysis. Our exploratory cascade experiment (Appendix) applies the same idea to judge routing and defers a formal cascade-level guarantee to future work.

3 Method

CARE-Judge converts an LLM judge f into a *selective evaluator* $(\hat{f}, \hat{c})$ that either outputs a judgment $\hat{f}(x)$ or abstains, based on a confidence measure $\hat{c}(x)$ and a risk-controlled threshold $\hat{\lambda}$. The framework has three components: (1) multi-signal protocol-stability feature extraction, (2) learned correctness calibration, and (3) empirical risk-controlled threshold selection with an exact Clopper–Pearson bound, validated by measured risk-violation rates rather than a new finite-sample theorem.

3.1 Protocol-Stability Features

Given a pairwise evaluation item $x = (q, a_1, a_2)$ consisting of a prompt q and two candidate responses, we issue multiple judge calls under semantically equivalent evaluation protocols and measure the stability of the resulting verdicts. Let $f(x; r, \tau)$ denote the judge's output (winner $\in \{A, B\}$ and confidence $\in [0, 1]$) given rubric r and sampling temperature τ.

Rubric Perturbation Stability. We define a set of $K = 3$ semantically equivalent rubric variants $\{r_1, \ldots, r_K\}$ that are intended to express the same evaluation criteria in different wording (e.g., "Prefer the more correct response" vs. "Choose the answer a careful evaluator would prefer"); all variants are listed verbatim in the Appendix. For each item, we collect verdicts $\{v_1, \ldots, v_K\}$ where $v_k = f(x; r_k, 0)$. We compute:

$$\text{rubric\_vote\_share} = \max_{w \in \{A,B\}} \frac{|\{k : v_k = w\}|}{K}, \quad (1)$$

$$\text{rubric\_flip} = \mathbb{1}[|\{v_1, \ldots, v_K\}| > 1], \quad (2)$$

where rubric_flip $= 1$ if any rubric variant produces a different winner. The intuition is that a verdict that changes under paraphrase is *protocol-dependent* and not robust to the stated specification. We avoid the stronger claim that it is "content-irrelevant," since a paraphrase can also shift the weight a judge places on competing criteria; a negative-control experiment with intentionally non-equivalent rubrics (Section 4.2) shows that on a mid-capability judge, non-equivalent rubrics produce $2.2\times$ higher flip rates than equivalent ones, so equivalent-rubric flips reflect genuine instability; on a near-random judge the two conditions are indistinguishable, consistent with the signal being uninformative below the competence threshold.

Position-Swap Consistency. We judge the item in both response orderings: (q, a_1, a_2) and (q, a_2, a_1). Let $v_{\text{orig}} = f(x; r_1, 0)$ and $v_{\text{swap}} = f(\text{swap}(x); r_1, 0)$, where swap reverses the response order. We map v_{swap} back to the original coordinate system and compute:

$$\text{swap\_consistency} = \mathbb{1}[\text{map}(v_{\text{swap}}) = v_{\text{orig}}]. \quad (3)$$

This probes positional bias: a judge that reverses its verdict under response reordering is unreliable.

Self-Consistency. We form a set of S judgments under the base rubric that pools the deterministic base verdict (τ=0) with $S - 1$ stochastic re-samples at temperature $\tau > 0$, and compute the majority vote share:

$$\text{self\_vote\_share} = \max_{w} \frac{|\{s : f(x; r_1, \tau_s) = w\}|}{S}. \quad (4)$$

In all reported experiments $S = 3$ (one τ=0 base verdict and two τ=0.7 samples). An optional adaptive early-stopping rule to reduce cost was disabled in the main experiments (Appendix C).

Feature Vector. CARE-Judge uses six raw judge calls per item—one base (τ=0), two self-consistency samples (τ=0.7), one position swap, and two additional rubric paraphrases (the base rubric doubling as the third rubric verdict)—and forms a 13-dimensional feature vector grouped into three families (full definitions in Appendix E, Table 5),

$$\begin{aligned}\boldsymbol{\phi}(x) = \big[\, &\underbrace{\text{swap\_consistency, swap\_conf\_gap,}}_{\text{protocol: position}} \\ &\underbrace{\text{rubric\_vote\_share, rubric\_entropy, rubric\_flip,}}_{\text{protocol: rubric}} \\ &\underbrace{\text{self\_vote\_share, self\_entropy,}}_{\text{protocol: self-consistency}} \\ &\underbrace{\text{confidence, base\_conf, mean\_conf, std\_conf,}}_{\text{confidence}} \\ &\underbrace{\text{length\_gap, score\_margin}}_{\text{length}} \,\big].\end{aligned} \quad (5)$$

The three protocol-stability families (position, rubric, self-consistency) are the contributed signal; the confidence and length families are standard baselines the calibrator may exploit where informative. We *exclude* the simulated-annotator signal from $\boldsymbol{\phi}$ (a disabled N=0 feature would only add a spurious degree of freedom) and instead report Trust-or-Escalate as a dedicated, leakage-controlled baseline in the head-to-head comparison (Section 4.3, Appendix C).

3.2 Correctness Calibration

We fit a calibrator $\hat{p}(x) = P(\hat{f}(x) = y \mid \boldsymbol{\phi}(x))$ predicting the probability that the judge's verdict is correct given the features (y the reference label), supporting logistic, isotonic, and gradient-boosted variants. It is fit on a *training split* D_{train} disjoint from both the threshold-selection and evaluation sets.

3.3 Empirical Risk-Targeted Threshold Calibration

Given a risk tolerance α and error level δ, we select an acceptance threshold $\hat{\lambda}$ on a *calibration split* D_{cal} (disjoint from D_{train}) targeting the calibration-conditional criterion

$$P\Big(P\Big(\hat{f}(x) \neq y \mid \hat{c}(x) \geq \hat{\lambda}\Big) \leq \alpha\Big) \geq 1 - \delta, \quad (6)$$

with the outer probability over the calibration draw. We select $\hat{\lambda}$ by an *empirical prefix search*: sort D_{cal} by descending confidence and, among prefixes of size $\geq m$, take the largest whose $(1-\delta)$ upper bound on accepted-set error is $\leq \alpha$. Because the acceptance set is chosen data-dependently this is not exact family-wise control; we treat Eq. (6) as a design target and validate it *empirically* via the realized violation frequency $\Pr_{\text{seed}}(\hat{R}_{\text{test}} > \alpha)$ over many held-out splits (Section 4). We use the *exact* one-sided Clopper–Pearson upper bound (Clopper and Pearson 1934):

$$\text{CP}_{\text{upper}}(e, n, \delta) = \sup\{p : P(\text{Bin}(n, p) \leq e) \geq \delta\}, \quad (7)$$

where e is the number of errors among the n accepted calibration items, computed via bisection on the binomial CDF. The complete train/calibrate/test procedure, including the minimum-acceptance floor m that prevents certifying on too few points, is given as Algorithm 1 in Appendix A. The selective evaluator is then applied to a *held-out test split* D_{test}:

$$(\hat{f}, \hat{c})(x) = \begin{cases} \hat{f}(x) & \text{if } \hat{c}(x) \geq \hat{\lambda}, \\ \emptyset & \text{otherwise (abstain).} \end{cases} \quad (8)$$

All reported metrics (coverage, selective risk, accepted accuracy, calibration error) are measured on the held-out D_{test}; the empirical violation rate over many splits is our primary evidence that Eq. (6) holds in practice. As an application, the same threshold rule extends to a cost-aware cheap-to-strong *cascade* with per-tier conditional calibration and a $\delta_t = \delta/T$ union-bound error budget; we treat it as exploratory and defer it to Appendix F.

4 Experiments

4.1 Experimental Setup

Datasets. We use four benchmarks spanning objective correctness and subjective preference: JudgeBench (Tan et al. 2024) (620 objectively-labeled pairs over knowledge/reasoning/math/coding), RewardBench (Lambert et al. 2024) (chosen/rejected pairs over chat/safety/reasoning), TL;DR (Stiennon et al. 2020) (summarization preference), and LMArena (Zheng et al. 2023) (Chatbot-Arena human preference). We scale each subjective benchmark to 2,000 pairs (dropping ties) so the 40/30/30 split gives calibration/test folds of several hundred items—enough for the exact Clopper–Pearson bound to certify nonzero coverage. We redistribute only derived feature traces, not raw text, under each benchmark's license.

Judge Models. We use five judges. Two API judges span the mid-to-frontier range: **DeepSeek-V4** in non-thinking `deepseek-chat` mode (mid-capability; the reasoning `v4-pro` mode is too token-intensive for six-call collection) and **GPT-5.5** (frontier, default reasoning effort). The other three form a *controlled* same-family ladder—**Qwen2.5-1.5B/7B/14B-Instruct**—holding recipe, tokenizer, and prompt fixed so that only parameter count varies, isolating a size-driven capability trend from the broader (confounded) API-vs-open comparison. Base confidence is verbalized confidence, since log-probabilities are not universally exposed. Exact snapshot identifiers, dates, decoding settings, and invalid-output handling are in Appendix C. (GLM-5.2 was attempted but behaves as a slow reasoning model on the available endpoint, so we omit it.)

Feature Collection, Protocol, and Baselines. Each item uses six judge calls: one base (τ=0), two self-consistency samples (τ=0.7, pooled to S=3), one position swap, and two rubric paraphrases (K=3 with the base rubric). Unparseable model outputs are retried once, then recorded as an `invalid` state that counts as a disagreement (lowering, never inflating, stability); genuine parse-invalid rates are $< 3\%$ everywhere. Separately, a subset of GPT-5.5 API calls failed with transient service errors (rate-limit/quota) rather than model outputs; these are not judge behavior, so the affected items are excluded from GPT-5.5's analysis and we report its per-benchmark retained N in Table 1 (Appendix C). Ties are dropped on the preference benchmarks. All results use a strict 40/30/30 train/calibration/test split (fit calibrator / select threshold / report), 20 seeds, α=0.15, δ=0.10, exact Clopper–Pearson. At the matched six-call budget we compare *Raw*, *CARE* (logistic fusion), the single signals *base*/*rubric*/*swap* (a budget-matched reimplementation of SCOPE's bidirectional signal (Badshah, Emami, and Sajjad 2026))/*self*, oracle *best-single*, and a budget-matched Trust-or-Escalate simulated-annotator baseline (Jung, Brahman, and Choi 2025).

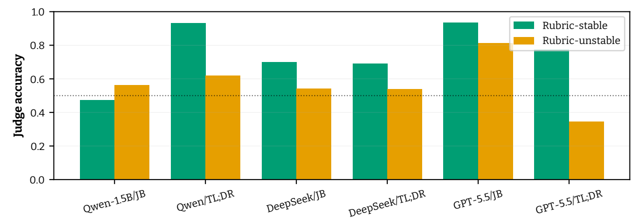

Figure 2: Judge accuracy on rubric-stable vs. rubric-unstable subsets. For mid-capability DeepSeek-V4 the stable subset is more accurate on every benchmark (up to $+27.4$ points on RewardBench); for frontier GPT-5.5 the effect shrinks or reverses, as its rare instabilities fall on ambiguous rather than wrong pairs (RQ1).

4.2 Main Results

Table 1 presents the main held-out results across all model–dataset combinations.

Finding 1 (RQ1): protocol *instability* predicts error for mid-capability judges. For DeepSeek-V4 (verdicts flip on 6–20% of items), rubric and position-swap instability separate lower-accuracy subsets on all four benchmarks (Fig. 2, Appendix I): the RewardBench gap is largest, with rubric-stable verdicts at 93.7% accuracy versus 66.4% when unstable ($+27.4$pp), and all DeepSeek gaps are positive ($+4.1$ to $+27.4$pp). The relationship is *capability-gated and can invert*: for the frontier GPT-5.5 (rubric-flip $<8\%$) unstable items are not systematically wrong, and on RewardBench the gap even reverses (-5.3pp rubric, -14.2pp position) as its rare disagreements fall on genuinely ambiguous pairs. The signal is thus a mid-capability warning, not a universal law; the learned calibrator (Finding 2) still recovers usable structure where a single raw gap is small or reversed.

Negative control. To rule out that equivalent-rubric flips are mere wording noise, we contrast the K=3 equivalent paraphrases against an *intentionally non-equivalent* rubric set (brevity/thoroughness/formality) on JudgeBench. On DeepSeek-V4 the non-equivalent set flips $2.2\times$ more often (28.0% vs. 12.8%), so equivalent-rubric flips are a conservative lower bound on real instability, not artifacts; the near-random Qwen-1.5B shows no such ordering (31.0% vs. 16.7%), being insensitive to what the rubric asks.

Finding 2 (RQ2): fusion helps in the mid-capability regime and not elsewhere. RQ2 asks not whether fusion helps *on average* across all judges—Finding 1 predicts it should not—but whether it helps in the mid-capability band where the signal is informative. Two independent judges occupy that band: DeepSeek-V4 (a frontier-lab API model) and Qwen2.5-7B (an open ladder model), which share no training recipe. On DeepSeek-V4, fusion improves over the best single signal by $+6.3$ AUROC on LMArena ($0.615 \to 0.678$),

Table 1: Held-out selective evaluation and matched-budget signal comparison (logistic fusion, 20 seeds, α=0.15, δ=0.10, exact Clopper–Pearson, strict 40/30/30 split). Judges are ordered by raw capability: below-threshold (Qwen2.5-1.5B), mid-capability (DeepSeek-V4, Qwen2.5-7B), and frontier (GPT-5.5). N is items (test fold $0.3N$); coverage/accepted-accuracy are pooled across seeds; "Viol." is the seed fraction with held-out accepted error above α; coverage 0.000 (Acc. "—") means no confidence level meets the risk constraint, so CARE abstains. The right block reports matched six-call-budget AUROC per single signal versus the **fused** calibrator (best per row **bold**); "swap" is the SCOPE-style bidirectional signal, "best-1" the oracle best single signal on train.

		Selective (α=0.15)				Matched-budget AUROC				
Model	Dataset	N	Cov.	Acc.	Viol.	base	swap	rubric	best-1	**fusion**
Qwen2.5-1.5B	JudgeBench	620	0.000	—	0.00	0.497	**0.534**	0.534	0.534	0.523
	TL;DR	2000	0.000	—	0.00	0.503	0.570	0.535	0.570	**0.583**
	RewardBench	2000	0.516	0.880	0.10	0.609	0.425	0.436	0.609	**0.713**
	LMArena	2000	0.000	—	0.00	0.515	0.563	0.553	0.563	**0.578**
DeepSeek-V4	JudgeBench	620	0.014	0.933	0.00	0.593	0.561	0.522	0.593	**0.603**
	TL;DR	2000	0.085	0.841	0.40	0.537	0.616	0.549	0.616	**0.675**
	RewardBench	2000	0.999	0.926	0.00	0.787	0.606	0.591	0.787	**0.831**
	LMArena	2000	0.077	0.894	0.05	0.604	0.615	0.562	0.615	**0.678**
Qwen2.5-7B	JudgeBench	619	0.000	—	0.00	0.510	0.578	0.511	0.578	**0.548**
	TL;DR	2000	0.000	—	0.00	0.549	0.616	0.515	0.616	**0.636**
	RewardBench	2000	0.987	0.877	0.00	0.611	0.605	0.566	0.611	**0.687**
	LMArena	1952	0.000	—	0.00	0.522	0.588	0.539	0.588	**0.599**
GPT-5.5	JudgeBench	620	0.994	0.915	0.00	**0.845**	0.695	0.576	**0.845**	0.834
	TL;DR	2000	0.218	0.838	0.20	0.650	0.629	0.537	0.650	**0.667**
	RewardBench	1999	0.997	0.961	0.00	**0.904**	0.757	0.546	**0.904**	0.888
	LMArena	1999	0.126	0.653	0.30	0.634	0.571	0.548	0.634	**0.645**

+5.9 on TL;DR (0.616 $\rightarrow$ 0.675), and +4.4 on RewardBench (0.787 $\rightarrow$ 0.831), with paired-bootstrap 95% CIs over the 20 seeds excluding zero on those three benchmarks (JudgeBench, at +1.0, includes zero). On Qwen2.5-7B fusion is positive on three of the four benchmarks (+7.6/+2.0/+1.1 AUROC on RB/TL;DR/LMArena over the best single signal) but falls 3.0 below the swap signal on the objective-label JudgeBench, where a 7B open judge is near chance. Seven of the eight mid-capability pairs thus show positive fusion gains, as competence-gating predicts.

On the frontier GPT-5.5 fusion ties its already-strong base confidence to within noise (e.g. 0.834 vs. 0.845 on JudgeBench and 0.888 vs. 0.904 on RewardBench, where base confidence edges fusion by ≤ 0.02), consistent with there being little instability left to exploit; below the competence threshold no signal exceeds ≈ 0.58 on Qwen-1.5B's subjective benchmarks.

The regime boundary is fixed *before* any fusion result is examined: RQ1 (Finding 1) shows instability tracks error only for judges of at least moderate raw accuracy, so we assign each judge to a regime by its raw agreement alone—mid-capability at 0.65–0.80 (DeepSeek-V4, Qwen2.5-7B), frontier above it (GPT-5.5), below-floor near chance (Qwen2.5-1.5B). The **pre-specified primary test** is therefore restricted to the eight mid-capability pairs where the mechanism is predicted to operate: fusion improves seven of them (+1.0 to +7.6 AUROC), the sole exception being Qwen2.5-7B on JudgeBench. As a *secondary* comparison, pooling all 17 judge–benchmark pairs—including regimes where fusion cannot help by construction—gives a one-sided Wilcoxon p=0.087 (11/17 wins); the direction of the test follows RQ1's directional prediction that fusion should not hurt. The gap between the mid-capability result and the marginal pooled one is what competence-gating predicts: pooling averages in the frontier and below-threshold regimes where no signal can help. In selective terms, at α=0.15 CARE raises accepted accuracy above raw by abstaining on the risky tail (e.g. DeepSeek-V4/TL;DR: 8.5% coverage at 84.1% accuracy, raw 67.9%), while a saturated judge already meeting the budget is accepted almost entirely (GPT-5.5/RewardBench 99.7% at 96.1%); the "Viol." column reports risk-control violations, expected occasionally at low coverage under the calibration-conditional target (Appendix H).

Finding 3 (RQ3): stability is not correctness—the competence boundary. Below a minimum judge capability the signal inverts: where Qwen2.5-1.5B is genuinely near chance (JudgeBench 0.49, TL;DR/LMArena 0.58) its stable verdicts are not more correct, so CARE-Judge accepts zero examples rather than trust them. RewardBench is the instructive exception: there the same 1.5B judge is far more accurate (raw 0.77) and fusion behaves as in the mid-capability regime (0.713 AUROC, 51.9% coverage at 88.0%). Competence is thus *per task, not per model*—the threshold tracks how accurate the judge actually is on the data at hand, as a raw-accuracy-indexed account predicts. We quantify the near-chance boundary in Section 4.4; safety (abstention) and usefulness (coverage) must be judged jointly.

4.3 Head-to-Head System Comparison

We compare CARE directly against the two prior selective-judging *systems*—SCOPE and Trust-or-Escalate—at the same six-call budget (full table in Appendix D). CARE attains the highest AUROC on all eight DeepSeek/GPT pairs. The margin over SCOPE's bidirectional-consistency score is large for DeepSeek-V4 (0.831 vs. 0.606 on RewardBench) and remains clear for GPT-5.5 (0.888 vs. 0.757 on RewardBench): a frontier judge rarely flips under reordering, so the consistency signal is only weakly informative, whereas fusion additionally leverages its well-calibrated base confidence. Trust-or-Escalate's agreement hovers near chance (0.50–0.53) at $N{=}0$ dedicated annotators. No single signal is uniformly best; the learned fusion is what delivers robustness across the capability spectrum.

4.4 Why Stability Is Not Correctness

We decompose Qwen-1.5B's 620 JudgeBench verdicts by correctness and rubric stability: 65% of its 314 incorrect verdicts are *rubric-stable-wrong* (203/314), and only 51.1% of its 415 rubric-stable verdicts are correct—essentially chance. This quantifies the competence boundary underpinning RQ3 and is consistent with mechanistic analyses of judge bias (Xu et al. 2026; Panickssery, Bowman, and Feng 2024): a shared heuristic yields highly consistent but wrong answers.

What the stable-but-wrong errors look like. These 203 items reflect a confidently held wrong heuristic, not noise: their mean verbalized confidence (0.903) is indistinguishable from that on *correct* verdicts (0.904), so confidence is useless exactly where a stability signal is meant to help—and it still fails. The errors are not the classic length bias (the judge picks the *shorter* response 79% of the time), while position instability is pervasive (73.9% swap-inconsistency vs. 71.1% overall). Below the floor the protocol signals do not merely weaken—they cease to co-vary with correctness, the operational content of "competence-gating."

4.5 Ensemble Baselines and Feature-Group Ablation

Against two ensemble baselines (majority/weighted vote) and a by-group feature ablation (full tables in Appendix J), CARE is best or tied-best in 9 of 12 pairs. Weighted vote is competitive for GPT-5.5 (it uses confidence implicitly) but has no abstention mechanism: a vote must rule on every item, whereas on DeepSeek-V4/TL;DR CARE abstains to 8.5% coverage at 84.1% accuracy versus the raw 67.9%. The group ablation confirms complementary roles: confidence dominates for GPT-5.5, protocol features carry the ladder judges, and length is near-chance throughout.

4.6 Judge-Capability Dependence: A Controlled Ladder

Is the signal useful because judges are *more capable* or merely *different systems*? A *controlled* same-family ladder (Qwen2.5-1.5B/7B/14B) on JudgeBench varies only parameter count (Table 2): raw accuracy rises monotonically with scale ($0.49 \rightarrow 0.57 \rightarrow 0.60$), while fused AUROC rises from 1.5B to 7B ($.52\pm.06 \rightarrow .55\pm.04$) and then does not increase at 14B ($.54\pm.04$). Since the 7B and 14B values differ by less than a seed-level standard deviation, we read the signal as *saturating by the 7B scale*, consistent with gains concentrated in the mid-capability transition rather than growing linearly with size. We restrict the ladder to JudgeBench's 620 objective-labeled items (sufficient to isolate the size effect at fixed recipe; the 14B judge runs at only $\approx$ 200 items/hour). The cross-system comparison agrees—on JudgeBench DeepSeek-V4 (raw 0.68) yields AUROC 0.60 and GPT-5.5 (raw 0.91) yields 0.83—but as the API judges differ in recipe we call that gap capability-suggestive, not controlled.

Table 2: Controlled capability ladder on JudgeBench: same-family Qwen2.5 judges varying only in parameter count (20 seeds, 620 items). Raw accuracy rises with scale; fused AUROC rises from 1.5B to 7B and then saturates (7B and 14B differ by less than one standard deviation).

Judge	Params	Mean raw acc.	Fused AUROC
Qwen2.5-1.5B	1.5B	0.49	$.52\pm.06$
Qwen2.5-7B	7B	0.57	$.55\pm.04$
Qwen2.5-14B	14B	0.60	$.54\pm.04$

The mid-capability replication (Qwen2.5-7B). The 7B rung doubles as the second mid-capability judge underpinning RQ2 (Section 4.2): sharing DeepSeek-V4's accuracy band but none of its recipe, it makes the fusion claim a recipe-independent replication, positive on three of its four benchmarks (the exception being the objective-label JudgeBench), reproducible via the released `scripts/run_e1_midcap.sh`.

4.7 Robustness and Calibration

Over 20 random splits on DeepSeek/TL;DR, fused AUROC is 0.675 ± 0.016 (CV 2.4%); GPT-5.5's fused scores are well calibrated (ECE $\leq$ 0.06) while DeepSeek-V4 is looser (ECE 0.16–0.29). Sweeping $\delta \in \{0.05, 0.10, 0.20\}$ and calibration-set size ($n_{\text{cal}} \in [200, 1000]$) leaves AUROC essentially unchanged (0.676 ± 0.001), so CARE-Judge does not require a large labeled calibration set.

5 Conclusion

We studied *protocol stability* as a per-instance reliability signal for LLM-as-a-judge and found it to be *competence-gated*. A verdict that changes under semantically equivalent rubrics, orderings, or resampling is unreliable—but only when the judge is capable enough that its instabilities track genuine difficulty; below a competence threshold the signal inverts and a near-random judge is stably wrong. CARE-Judge operationalizes this by fusing the three instability families into a learned calibrator with inherited selective-risk control (Jung, Brahman, and Choi 2025; Badshah, Emami, and Sajjad 2026). The same mechanism accounts for fusion helping in the mid-capability band (DeepSeek-V4 and Qwen2.5-7B), tying a saturated frontier judge, and inverting below the threshold; fusion is positive on seven of the eight pre-specified mid-capability pairs, while the pooled average is diluted by regimes where fusion cannot help.

Practical recommendation. A simple rule follows: deploy protocol-stability abstention for mid-capability judges (roughly 0.65–0.8 raw agreement), where fusion buys the largest risk-controlled coverage; for a frontier judge with near-perfect verbalized confidence the extra perturbation calls are optional; and below the competence threshold, abstain by default rather than trust stable verdicts.

Limitations and Future Work. Our risk control is empirical (a data-dependent threshold validated by measured violation rates, not a family-wise-corrected guarantee) and requires reference labels; stability does not imply correctness, and a shared-bias judge can defeat every signal at once. A crossed size$\times$recipe study, a conditionally-calibrated cascade with a formal guarantee, and extension to pointwise scoring are future work. Because abstention is not neutral—abstaining disproportionately on certain slices could skew downstream evaluation—deployers should audit *which* items are abstained on, not just aggregate coverage.

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Table 3: Alpha-sweep risk-coverage tradeoff (logistic fusion, 5 seeds—fewer than the 20 seeds used for the main tables because this sweep spans six α levels, and the qualitative trend is stable across seeds—*pooled* across seeds). Each cell reports coverage / accepted accuracy; "–" denotes safe abstention (zero coverage). Coverage is *positively correlated* with the risk budget α but not strictly monotone: because the accepted prefix is chosen data-dependently on each split, relaxing α occasionally selects a different threshold that lowers pooled coverage. We report the trend as-is rather than enforcing monotonicity.

Model	Dataset	α=0.05	α=0.10	α=0.15	α=0.20	α=0.25	α=0.30
DeepSeek-V4	JudgeBench	–/–	.03/.96	.03/.93	.03/.93	.24/.78	.46/.71
	TL;DR	–/–	–/–	.09/.84	.13/.85	.27/.79	.67/.73
	RewardBench	.82/.97	1.00/.93	1.00/.93	1.00/.93	1.00/.93	1.00/.93
	LMArena	–/–	–/–	.07/.89	.05/.90	.63/.78	.97/.73
GPT-5.5	JudgeBench	–/–	.25/.94	.32/.90	.39/.88	.42/.82	.44/.79
	TL;DR	–/–	.02/.96	.06/.98	.18/.87	.59/.79	.73/.74
	RewardBench	.73/.96	.79/.92	.82/.88	.86/.84	.91/.79	.97/.74
	LMArena	–/–	–/–	.05/.91	.09/.88	.39/.79	.63/.75

Supplementary Material

Reviewers are not obligated to read this material (AAAI-27 policy).

A CARE-Judge Algorithm

Algorithm 1 CARE-Judge selective evaluation

Input: labeled items D; risk α, level δ; min-accept m
Split D into disjoint $D_{\text{train}}, D_{\text{cal}}, D_{\text{test}}$ (40/30/30)
Fit calibrator $\hat{p}$ on $\{(\boldsymbol{\phi}(x), \mathbb{1}[\hat{f}(x)=y])\}_{x \in D_{\text{train}}}$
Sort D_{cal} by $\hat{p}$ descending: $c_{(1)} \geq \cdots \geq c_{(n)}$
$\hat{\lambda} \leftarrow \infty$ {accept nothing by default}
for $i = m$ **to** n **do**
    $e_i \leftarrow \#\{j \leq i : \hat{f}(x_{(j)}) \neq y_{(j)}\}$ {errors in top-i prefix}
    if $\text{CP}_{\text{upper}}(e_i, i, \delta) \leq \alpha$ **then**
        $\hat{\lambda} \leftarrow \hat{p}(x_{(i)})$ {largest certified prefix so far}
    end if
end for
Output: on D_{test}, accept x iff $\hat{p}(x) \geq \hat{\lambda}$, else abstain
Report: empirical violation rate $\Pr_{\text{seed}}(\hat{R}_{\text{test}} > \alpha)$ over splits

B Alpha-Sweep Risk–Coverage Tradeoff

C Reproducibility Details

Judge model configurations. GPT-5.5 was queried through an OpenAI-compatible endpoint at its default reasoning effort; DeepSeek-V4 was queried in non-thinking (`deepseek-chat`) mode because the reasoning `v4-pro` mode is too token-intensive for six-call-per-item feature collection. The controlled capability ladder uses `Qwen/Qwen2.5-1.5B-Instruct`, `Qwen/Qwen2.5-7B-Instruct`, and `Qwen/Qwen2.5-14B-Instruct` run locally in fp16 on a single 24 GB GPU, with identical prompts, chat template, and decoding settings so that only parameter count varies. All deterministic verdicts use temperature 0; self-consistency samples use temperature 0.7 with `max_new_tokens`$= 80$. Base confidence is the model's verbalized confidence, since token log-probabilities are not exposed by all endpoints. We pin and release the exact API model snapshot identifiers and dates, the local model commit hashes, all prompt templates, the six-call collection code, per-seed split indices, and the fitted calibrators; the AAAI reproducibility checklist is completed in the supplementary archive.

Compute budget. All local judges (the Qwen2.5 ladder) were run on a single 32 GB consumer GPU (RTX 5090). The 1.5B judge collects features at ≈1,500 items/hour, the 7B at ≈450/hour, and the 14B at ≈200/hour; total local feature collection for the ladder was under 40 GPU-hours. Each item consumes exactly six judge calls, so feature collection for the API judges required $6 \times 6{,}620 \approx 40{,}000$ calls per API model (DeepSeek-V4 and GPT-5.5), plus 2×898 additional calls for the DeepSeek negative-control. The calibrator fit and all 20-seed analyses run in minutes on a CPU. This modest budget is intentional: CARE-Judge is designed to be reproducible on a single commodity GPU plus metered API access.

Tie and invalid handling. An output that cannot be parsed as A/B is retried once. If it is still unparseable we record it as an explicit `invalid` state rather than silently mapping it to the base verdict: an `invalid` perturbation call counts as a *disagreement* with the base verdict, so it *lowers* the corresponding stability signal and can only push an item toward abstention. We deliberately avoid the tempting shortcut of mapping invalid→base verdict, because that would artificially inflate measured stability (an unparseable output would be counted as agreement). Genuine parse-invalid rates (the model returned an output we could not map to A/B) are below 3% for every judge–benchmark pair. We distinguish these from *transient API-service failures* (rate-limit/quota responses that return no model output): during GPT-5.5 collection a subset of calls failed this way, and because such failures reflect infrastructure rather than judge behavior we *exclude* the affected items from GPT-5.5's analysis rather than count them as verdicts. The retained per-benchmark sample sizes for GPT-5.5 are JudgeBench 620, TL;DR 2,000, RewardBench 1,999, and LMArena 1,999 (Table 1); all other judges retain their full sets. For the preference benchmarks (TL;DR, LMArena) genuine ties in the human label are dropped before evaluation; JudgeBench and RewardBench carry objective correct/incorrect labels and contain no ties.

Rubric variants and the negative control (full detail). The $K{=}3$ rubric paraphrases used for the rubric-stability signal are reproduced verbatim in the released prompt file. They were written to hold the evaluation target (which response is better) fixed while varying wording. The negative-control experiment summarized in Section 4.2 used an *intentionally non-equivalent* rubric set (one emphasizing brevity, one thoroughness, one formality) on both a mid-capability judge (DeepSeek-V4) and a near-random judge (Qwen2.5-1.5B), both on JudgeBench. The full per-condition counts are: **DeepSeek-V4**—equivalent set flip rate 12.8% (46/359) vs. non-equivalent 28.0% (151/539), a 2.2× ratio; **Qwen2.5-1.5B**—equivalent 31.0% (189/610) vs. non-equivalent 16.7% (102/610), i.e. the non-equivalent set was if anything *more* stable. The mid-capability judge moves when the criterion genuinely changes (so equivalent-rubric flips are a conservative floor on real instability), whereas the near-random judge latches onto surface features regardless of rubric content. This closes the loop on the competence-gated asymmetry.

D Head-to-Head System Comparison (Full Table)

Table 4 reports the full three-way *system* comparison summarized in Section 4.3, at the matched six-call budget.

Table 4: Three-way system comparison: held-out AUROC at the matched six-call budget (20 seeds, strict 40/30/30 split). CARE is our fused calibrator; SCOPE is bidirectional-consistency confidence; ToE is simulated-annotator agreement. Best per column in **bold**. Against full systems (not isolated raw signals) CARE wins all eight pairs.

	DeepSeek-V4				GPT-5.5			
Method	JB	TL;DR	RB	Arena	JB	TL;DR	RB	Arena
CARE (fusion)	**.603**	**.675**	**.831**	**.678**	**.834**	**.667**	**.888**	**.645**
SCOPE (bidir.)	.561	.616	.606	.615	.695	.629	.757	.571
ToE (sim. agr.)	.510	.519	.513	.517	.526	.511	.499	.510

E Feature Definitions

Table 5 defines all 13 features of the vector $\boldsymbol{\phi}(x)$ in Eq. (5). Confidence values are the judge's verbalized self-reported confidence in $[0,1]$; entropies are computed over the empirical winner distribution across the relevant call set and normalized to $[0,1]$.

Per-seed risk reporting. For every model–dataset–α setting we release, per seed: the accepted count, coverage, realized test error $\hat{R}_{\text{test}}$, the Clopper–Pearson upper bound on the calibration split, and whether $\hat{R}_{\text{test}} > \alpha$. To characterize variance rather than only central tendency, we resample the 40/30/30 split over many random seeds—up to several hundred paired splits per setting where feasible—and report coverage and selective risk as mean±std across these splits alongside the pooled point estimates used in the main tables (pooling errors and accepts, rather than averaging per-seed ratios, so that zero-coverage seeds neither dominate nor are silently dropped). Aggregate seed robustness is high (coverage coefficient of variation $\approx 2.4\%$ at $\alpha{=}0.15$). We also report the empirical risk-violation frequency $\Pr_{\text{seed}}(\hat{R}_{\text{test}} > \alpha)$ as our direct check on risk control; because the acceptance threshold is chosen data-dependently, this empirical frequency—not a formal family-wise guarantee—is the quantity we ask readers to judge, and occasional violations at very low coverage (where only a handful of items are accepted) are expected and reported transparently.

F Exploratory: Cost-Aware Cascade

This section uses the **same strict 40/30/30 split as the main experiments** and calibrates each tier *conditionally* on the calibration items that actually reach it, with error budget $\delta_t = \delta/T$ (Section 3.3). We still label it exploratory because it uses a single labeled pool per dataset and we have not stress-tested the conditional calibration under distribution shift.

We evaluate a three-tier cascade (Qwen-1.5B → DeepSeek-V4 → GPT-5.5), averaging over 10 seeds (Table 6). On RewardBench the cascade is highly effective: Qwen and DeepSeek certify most items, achieving 91–99% paid-API savings at near-full

Table 5: The 13 features of $\phi(x)$ (Eq. 5), grouped by family.

Feature	Definition
Protocol: position (swap)	
swap_consistency	$\mathbb{1}$[verdict unchanged after response reorder] (Eq. 3).
swap_conf_gap	$\lvert \mathrm{conf}_{\mathrm{orig}} - \mathrm{conf}_{\mathrm{swap}} \rvert$: confidence change under reorder.
Protocol: rubric	
rubric_vote_share	majority winner share over K=3 rubric paraphrases (Eq. 1).
rubric_entropy	normalized entropy of the winner distribution over the K rubrics.
rubric_flip	$\mathbb{1}$[any rubric paraphrase changes the winner] (Eq. 2).
Protocol: self-consistency	
self_vote_share	majority winner share over S=3 samples (Eq. 4).
self_entropy	normalized entropy of the winner distribution over the S samples.
Confidence	
confidence	aggregate verbalized confidence pooled across the perturbation calls.
base_conf	verbalized confidence of the deterministic base verdict (τ=0).
mean_conf	mean verbalized confidence across all six calls.
std_conf	standard deviation of verbalized confidence across the six calls.
Length	
length_gap	normalized token-length difference $\lvert a_1 \rvert - \lvert a_2 \rvert$ between the two responses.
score_margin	difference between the base verdict's confidence and 0.5 (decisiveness).

Table 6: Cost-aware cascade (strict 40/30/30 split, conditional per-tier calibration, δ_t=δ/T, 10 seeds). Cov/Acc on held-out test; Save is paid-API-token cost avoided vs. single-call GPT-5.5.

Dataset	α	Cov.	Acc.	Save
JudgeBench	0.15	0.258	0.933	−11%
	0.30	0.538	0.799	11%
TL;DR	0.15	0.051	0.935	−11%
	0.30	0.552	0.806	16%
RewardBench	0.15	0.998	0.914	91%
	0.30	0.998	0.780	99%
LMArena	0.15	0.045	0.869	−7%
	0.30	0.597	0.794	46%

coverage. On the subjective benchmarks (TL;DR, LMArena) and JudgeBench, the cheap tiers cannot certify the risk bound at strict α, so items escalate to GPT-5.5 and savings are near zero or negative; savings become positive only at loose α=0.30 where DeepSeek begins to certify. Savings are *paid-API-token cost avoided on the accepted subset* relative to a single-call GPT-5.5 judge, and exclude local GPU time, throughput, and latency.

G Exploratory: Cross-Dataset Transfer

We use the strict three-way split: the calibrator is fit on the *source* dataset (train+cal) and evaluated on a held-out *target* test split, averaged over 10 seeds (Table 7, GPT-5.5, α=0.15). The pattern is clear: calibrators trained on *subjective* benchmarks (LMArena, TL;DR) transfer with very high precision but low coverage—an LMArena-trained calibrator accepts only 3.9% of JudgeBench but at 100% accuracy—while calibrators trained on *objective* benchmarks (JudgeBench, RewardBench) transfer with higher coverage but lower accuracy (65.6–69.8%). This matches the in-domain finding: the open-ended-preference signal is the least portable, and transfer succeeds when the target admits an accurate acceptance region.

H Risk–Coverage Curves

I Signal-Level Accuracy Gaps

The per-benchmark accuracy gaps summarized by Figure 2 (in the main body) are tabulated below.

J Ensemble Baselines and Feature-Group Ablation

Table 7: Cross-dataset transfer for GPT-5.5 (α=0.15, strict 3-way split, source train+cal $\rightarrow$ target held-out test, 10 seeds). Subjective sources (LMArena, TL;DR) yield precise but low-coverage transfer; objective sources (JudgeBench, RewardBench) yield higher coverage at lower accuracy.

Source $\rightarrow$ Target	Cov.	Acc.
TL;DR $\rightarrow$ JudgeBench	0.107	0.995
TL;DR $\rightarrow$ RewardBench	0.536	0.986
TL;DR $\rightarrow$ LMArena	0.174	0.839
JudgeBench $\rightarrow$ TL;DR	0.539	0.756
JudgeBench $\rightarrow$ RewardBench	0.682	0.958
JudgeBench $\rightarrow$ LMArena	0.555	0.745
RewardBench $\rightarrow$ JudgeBench	0.558	0.656
RewardBench $\rightarrow$ TL;DR	0.789	0.697
RewardBench $\rightarrow$ LMArena	0.769	0.698
LMArena $\rightarrow$ JudgeBench	0.039	1.000
LMArena $\rightarrow$ TL;DR	0.006	1.000
LMArena $\rightarrow$ RewardBench	0.228	0.998

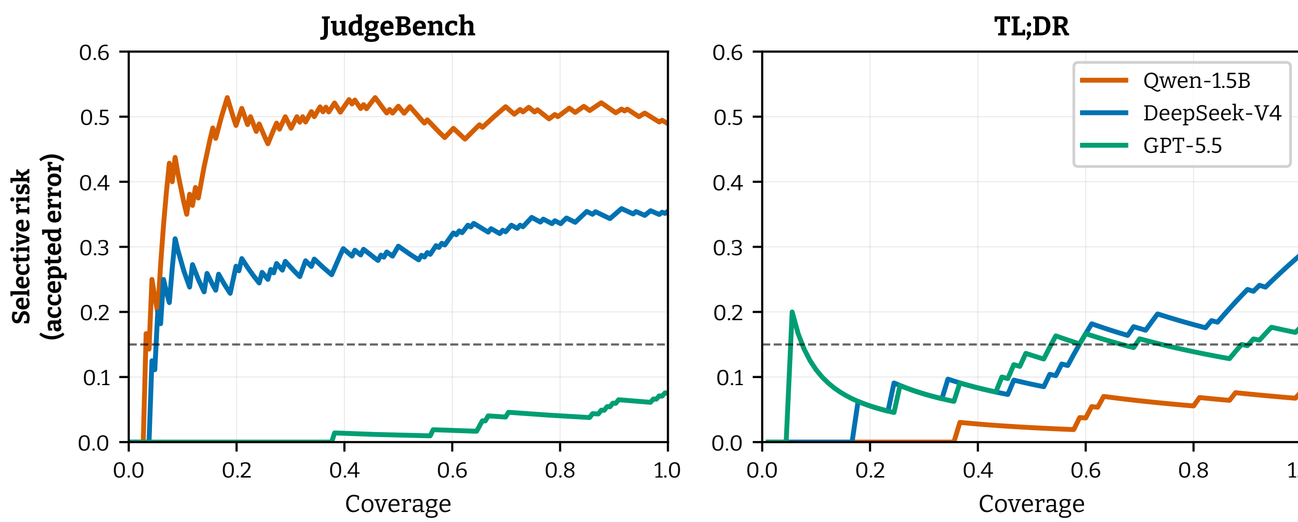

Figure 3: Held-out risk–coverage curves (logistic fusion). Lower is better; the dashed line marks the α=0.15 risk budget. GPT-5.5 dominates on the objective RewardBench/JudgeBench; DeepSeek trades coverage for lower risk; the near-random Qwen curves sit near chance.

Table 8: Signal-level accuracy: stable vs. unstable subsets, for judge–benchmark pairs with $\geq$60% raw accuracy. DeepSeek-V4 shows consistent positive gaps; GPT-5.5's rare instabilities do not track error (RewardBench).

Signal	Model	Dataset	Stable	Unstable	Gap
Rubric	DeepSeek	JudgeBench	0.689	0.648	+0.041
	DeepSeek	TL;DR	0.708	0.555	+0.153
	DeepSeek	RewardBench	0.937	0.664	+0.274
	DeepSeek	LMArena	0.749	0.478	+0.271
	GPT-5.5	RewardBench	0.709	0.762	−0.053
Position	DeepSeek	JudgeBench	0.728	0.575	+0.153
	DeepSeek	TL;DR	0.746	0.518	+0.228
	DeepSeek	RewardBench	0.929	0.867	+0.062
	DeepSeek	LMArena	0.772	0.528	+0.244
	GPT-5.5	RewardBench	0.708	0.850	−0.142

Table 9: Ensemble baselines vs. CARE (held-out AUROC, 20 seeds). CARE wins or ties in 9/12 pairs; weighted vote is competitive for GPT-5.5 (uses confidence implicitly) but lacks risk control.

Model	Data	**CARE**	SCOPE	MajVote	WtdVote
DeepSeek	JB	**0.603**	0.561	0.585	0.581
	TL;DR	**0.675**	0.616	0.609	0.618
	RB	**0.831**	0.606	0.675	0.810
	Arena	**0.678**	0.615	0.612	0.616
GPT-5.5	JB	**0.834**	0.695	0.583	0.806
	TL;DR	0.667	0.629	0.570	**0.673**
	RB	0.888	0.757	0.600	**0.899**
	Arena	**0.645**	0.571	0.568	0.643
Qwen	JB	0.523	**0.534**	0.505	0.509
	TL;DR	**0.583**	0.570	0.544	0.494
	RB	**0.713**	0.425	0.538	0.620
	Arena	**0.578**	0.563	0.559	0.494

Table 10: Feature-group ablation (held-out AUROC, 20 seeds). "FULL" = all 13 features; each ablation uses only that group. Confidence dominates for GPT-5.5; protocol for Qwen; length is near-chance everywhere.

Model	Data	**FULL**	protocol	conf	length
DeepSeek	JB	**0.603**	0.587	0.623	0.509
	TL;DR	0.675	0.660	**0.683**	0.548
	RB	**0.831**	0.699	0.828	0.622
	Arena	**0.678**	0.654	0.680	0.531
GPT-5.5	JB	0.834	0.692	**0.847**	0.510
	TL;DR	0.667	0.636	**0.673**	0.519
	RB	0.888	0.782	**0.906**	0.669
	Arena	0.645	0.598	**0.647**	0.505
Qwen	JB	0.523	**0.535**	0.494	0.497
	TL;DR	**0.583**	0.580	0.581	0.536
	RB	**0.713**	0.659	0.631	0.609
	Arena	0.578	0.579	**0.587**	0.535